

Brick by Brick: 75 Years of LEGO's Data-Driven Transformation

A semantic model and DAX-driven analysis of the full Rebrickable LEGO catalog, 1949–2024 — reframed as a story of platform growth and, more surprisingly, LEGO's quiet transformation into a lifestyle-merchandise brand.

[Lucrecia Krause](#) · [Business and Data Analysis Portfolio](#) · [Microsoft MVP, Power BI & Fabric](#)

Tool: Power BI · DAX

Scope: 1949–2024

Role: Data Analyst

Theme: Brick by Brick 2026

Original model & theme design

27,069

TOTAL CATALOG RECORDS

72%

ARE REAL BRICK SETS

149

THEMES TRACKED

1999

LICENSE INFLECTION YEAR

 This model was published in the **Microsoft Power BI Data Visualization World Championships** (Round 2, World Champs BCN 26), Contests Gallery, by Lucrecia Krause, MVP — [view the submission on the Microsoft Fabric Community](#).

01 • Overview

Why 75 years of LEGO data tells a business story

Most LEGO retrospectives are nostalgia pieces. This analysis treats LEGO's full public catalog — 27,069 records across 149 themes, sourced from the Rebrickable LEGO Database — as a case study in platform growth: how a single, unchanged core system, the interlocking brick, supported decades of expansion without losing structural coherence.

The core historical analysis covers 1949–2024 (75 years); the underlying date dimension is modeled through 2027 to support forward-looking, year-by-year exploration. A "top theme families" rollup (Theme Level 1 in the model) is used for the ranked/ribbon views elsewhere in this report — that curated view totals 7,281 sets across 20 families, a readable subset of the full 149-theme catalog, not a different dataset.

WHAT SURPRISED ME

LEGO grew from 5 sets in 1949 to 1,339 sets in a single year by 2025 — impressive by any measure. But digging into the data revealed something unexpected: 7,444 of 27,069 records, nearly 1 in 3, are not LEGO brick sets at all. They're keychains, backpacks, pencil cases, watches, DVD sets, and plush toys.

Key takeaway: LEGO's biggest transformation isn't the jump from 5 to 27,069 records — it's that 1 in 3 of those records are keychains, bags, and merchandise, not bricks. In 75+ years, LEGO became a lifestyle brand.

LEGO® Theme Lifespan - Born, Peaked & Gone

WHAT SURPRISED ME

LEGO grew from 5 sets in 1949 to 1,339 sets in a single year by 2025 — impressive by any measure. But digging into the data revealed something unexpected: 7,444 of 27,069 records — nearly 1 in 3 — are not LEGO brick sets at all. They are keychains, backpacks, pencil cases, watches, DVD sets and plush toys. The toy company we grew up with has become something else entirely.

WHAT PATTERN I DISCOVERED

Of 143 themes launched since 1949, 87 — that is 61% — have been discontinued. Themes rise fast and fall faster. But while brick themes come and go, the merchandise never stops growing. 835 keychain records alone nearly match Star Wars' entire 26-year run of 1,094 sets. The pattern is clear: LEGO experiments with brick themes, but bets long-term on merchandise.

KEY TAKEAWAY

LEGO's biggest transformation is not the jump from 5 to 27,069 records — it is that 1 in 3 of those records are keychains, bags and merchandise, not bricks. In 78 years LEGO became a lifestyle brand.

Source: Rebrickable LEGO Database (rebrickable.com)

27,069
TOTAL RECORDS

19,625
REAL BRICK SETS

7,444
NON-BRICK (28%)

835
KEYCHAINS ALONE

THE 7,444 NON-BRICK GEAR RECORDS — WHAT ARE THEY?

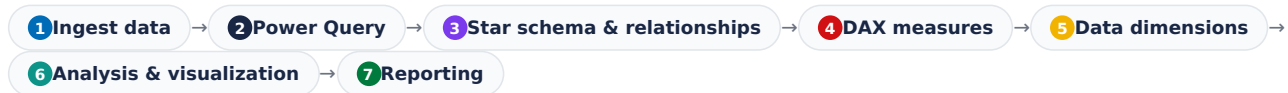


"LEGO® Theme Lifespan — Born, Peaked & Gone." Of 143 themes launched since 1949, 87 (61%) have been discontinued. The 7,444 non-brick gear records break down across categories like Key Chain, Bags/Totes & Luggage, Stationery, and Houseware — 835 keychain records alone nearly match Star Wars' entire 26-year run of 1,094 sets. Source: Rebrickable LEGO Database.

02 · Approach

The analyst workflow behind this report

This project follows a standard data-analyst pipeline end to end — every step below was done by me, in this order, on the same dataset:



1. Ingest — raw CSVs pulled from the Rebrickable LEGO Database (sets, themes, parts, colors, inventories, minifigures, part categories).

2. Power Query — each source table cleaned, typed, and reshaped — including one table (`inventories`) that had to be grouped and ranked to isolate the latest version of every set. Full step-by-step detail is in Section 03.

3. Star schema & relationships — a single fact table (`sets`) connected to dimension and bridge tables through explicit one-to-many relationships — no ambiguous or bidirectional joins.

4. DAX measures — eight measures built on top of the model, from a simple row count up to a safely-divided average — covered in full in Section 03.

5. Data dimensions — a fully custom `Dim_date` table (Year, Decade, Era, milestone flags) authored entirely in Power Query, with no source file, purpose-built to drive this report's storytelling.

6. Analysis & visualization — the ribbon, trend, ranked, and lifecycle charts covered in Section 04, plus the interactive year-by-year Time Capsule explorer.

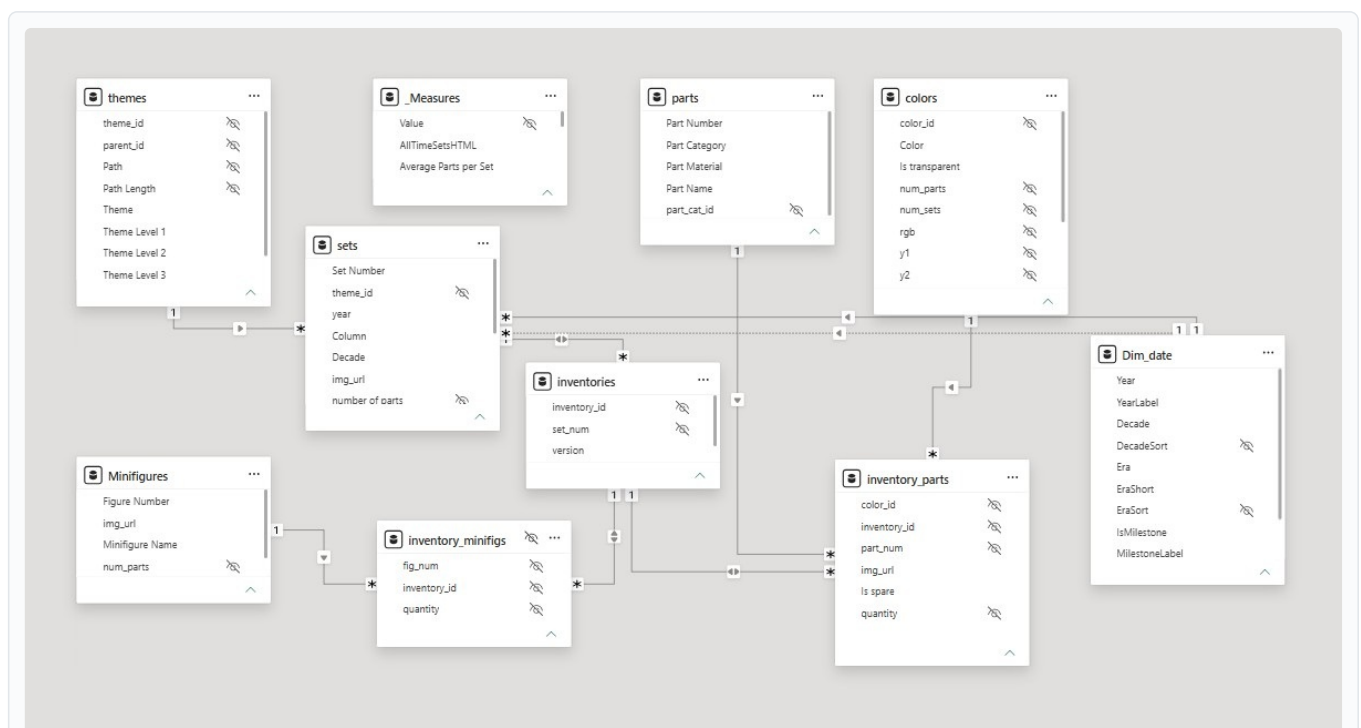
7. Reporting — this document itself, plus the published Power BI report — see the World Championship submission below.

The semantic model, DAX measures, and this report's custom Power BI theme (*Brick by Brick 2026*) were designed and built by me as part of this analysis — not a default template.

03 · Data Model

Semantic model & DAX relationships

The model is a star schema built around the **sets** fact table (Set Number, theme_id, year, Decade, number of parts), joined to dimension tables for **themes** (with a 3-level Theme Level 1/2/3 hierarchy and path), **colors**, and **Dim_date** (Year, Decade, Era, and milestone flags used to drive the turning-point callouts). Bridge tables — **inventories**, **inventory_parts**, and **inventory_minifigs** — connect sets to individual **parts** and **Minifigures**, enabling part- and minifigure-level analysis alongside set-level trends. A dedicated **_Measures** table holds the DAX layer, including *Average Parts per Set* and an HTML-formatted all-time-sets measure used for dynamic tooltips.



Semantic model — star schema with themes, sets, parts, colors, inventories, minifigures, date dimension, and the DAX measures table.

Manage relationships

+ New relationship

Autodetect

Edit

Delete

Filter

☐ From: table (column) ↑	Relationship	To: table (column)	Status	
☐ inventories (set_num)		sets (Set Number)	Active	...
☐ inventory_minifigs (fig_num)		Minifigures (Figure Number)	Active	...
☐ inventory_minifigs (inventory_id)		inventories (inventory_id)	Active	...
☐ inventory_parts (color_id)		colors (color_id)	Active	...
☐ inventory_parts (inventory_id)		inventories (inventory_id)	Active	...
☐ inventory_parts (part_num)		parts (Part Number)	Active	...
☐ sets (theme_id)		themes (theme_id)	Active	...
☐ sets (year)		Dim_date (Year)	Active	...
☐ sets (year)		Dim_date (YearLabel)	Inactive	...

Manage Relationships view — one-to-many joins linking the bridge tables (inventories, inventory_parts, inventory_minifigs) to their dimensions; sets(year) relates to Dim_date on two columns, with YearLabel kept inactive to avoid ambiguity.

POWER QUERY: TURNING RAW CSVS INTO A CLEAN MODEL

Every table below started as a plain Rebrickable CSV export. None of them were model-ready — the real work was in Power Query, deciding what to rename, what to reshape, and in one case, what to reconstruct entirely so a single DAX filter would work correctly.

inventory_sets

SHARED QUERY

1. Load CSV (3 columns, encoding 1252)

2. Promote headers

3. Type columns: `inventory_id` → Int64, `set_num` → text, `quantity` → Int64

Built once as a shared expression so any query needing set-level inventory counts can reference it without repeating the load logic.

part_categories

SHARED QUERY

1. Load CSV (2 columns)

2. Promote headers

3. Type columns: `id` → Int64, `name` → text

4. Rename: `id` → `part_cat_id`, `name` → `Part Category`

A small lookup table, shared so **parts** can merge it in cleanly rather than duplicating category logic.

inventories

POWER QUERY STEP · MOST COMPLEX TRANSFORMATION

1. Load CSV → promote headers → type columns

2. Rename `id` → `inventory_id`

3. Group by `set_num` — within each group, sort versions descending and add an index column called `version rev` (1 = latest version)

4. Expand the grouped table back out

This is the one table that isn't just cleaned, it's restructured. The group-and-rank step is what makes `inventories[version rev] = 1` possible — and that single column is what lets the **Parts** measure filter out duplicate/re-issued inventories instead of over-counting them.

sets

POWER QUERY STEP

1. Load CSV → promote headers → type columns

2. Rename `name` → `Set`, `num_parts` → `number of parts`

3. Add `Decade` column: `Number.IntegerDivide([year], 10) * 10 & "s"` (e.g. `1980s`)

4. Rename `set_num` → `Set Number`

The `Decade` column, built here rather than in DAX, is what powers every decade-level chart in this report — including the ribbon chart and the three-act trend view.

themes

POWER QUERY STEP

1. Load CSV → promote headers → type columns

2. Rename `id` → `theme_id`, `name` → `Theme`

Simple on the surface — the real hierarchy (Theme Level 1/2/3, parent/path) is built downstream in the model, not here.

parts

POWER QUERY STEP

1. Load CSV → promote headers → type columns
2. Merge with **part_categories** shared expression (Left Outer on `part_cat_id`)
3. Expand `Part Category` from the merged table
4. Rename: `part_material` → `Part Material`, `name` → `Part Name`, `part_num` → `Part Number`

This is where the shared **part_categories** expression actually earns its keep — merged in once, expanded, done.

colors

POWER QUERY STEP

1. Load CSV (8 columns) → promote headers → type all columns
2. Rename: `id` → `color_id`, `name` → `Color`, `is_trans` → `Is transparent`

Straightforward cleanup — but `Is transparent` is what lets the color analysis distinguish solid bricks from the transparent/trans-clear pieces.

inventory_parts

POWER QUERY STEP

1. Load CSV → promote headers → type columns
2. Rename `is_spare` → `Is spare`

The biggest table in the model by row count, and the lightest touch — it didn't need reshaping, just a readable column name.

inventory_minifigs

POWER QUERY STEP

1. Load CSV → promote headers → type columns (no renames)

Clean enough as-is — sometimes the right transformation is knowing when to stop.

Minifigures

POWER QUERY STEP

1. Load CSV → promote headers → type columns
2. Rename `name` → `Minifigure Name` , `fig_num` → `Figure Number`

Feeds the minifigure gallery in the year-by-year Time Capsule view.

Dim_date

GENERATED DIMENSION — NO CSV

1. Build year list 1949-2024 via `List.Numbers`
2. Convert to table, set `Year` → `Int64`
3. Add `Decade` (`1980s` format)
4. Add `DecadeSort` (integer, for correct visual ordering)
5. Add `Era` (7 LEGO historical periods)
6. Add `EraSort` (1-7 integer, for ordering)
7. Add `EraShort` (abbreviated era name)
8. Add `IsMilestone` (boolean — 14 key years flagged)
9. Add `MilestoneLabel` (text label for milestone years)
10. Add `YearLabel` (text version of Year, for display)
11. Add `YearsAgo` (`2024 - [Year]`)
12. Reorder columns to final output order

The only table with no source file at all — every column is authored in Power Query. This is what makes the turning-point callouts (1958, 1978, 1999, 2003, 2014, 2021) and the Era-based storytelling possible without a single hardcoded value in a visual.

KEY DAX MEASURES

Sets

#,0

```
Sets = COUNTROWS (sets)
```

Count of sets, version 1 only, implied by filter context. This is the base measure every ranked/ribbon chart in this report builds on.

Parts

#,0

```
Parts =  
    CALCULATE (  
        SUM (inventory_parts[quantity]),  
        inventories[version rev] = 1  
    )
```

Total part quantity, filtered to version-1 inventories only — avoids double-counting parts across re-released or corrected inventory versions.

Minifigures

#,0

```
Minifigures = SUM (inventory_minifigs[quantity])
```

Total minifigure quantity across all inventories — feeds the 19,082 minifigures KPI in the Time Capsule view.

Versions

0

```
Versions = COUNTROWS (inventories)
```

Count of all inventory versions, not just v1 — used to audit how many sets were revised/re-issued.

'Parts all'

HIDDEN · #,0

```
'Parts all' = SUM (inventory_parts[quantity])
```

All parts regardless of version. Kept hidden from report viewers — used internally as a building block for other measures, not for direct display.

'Unique Parts'

#,0

```
'Unique Parts' = DISTINCTCOUNTNOBLANK (inventory_parts[part_num])
```

Distinct part numbers, not quantity — answers "how many different parts exist," not "how many parts were produced."

'Unique Minifigures'

#,0

```
'Unique Minifigures' = DISTINCTCOUNTNOBLANK (inventory_minifigs[fig_num])
```

Distinct minifigure IDs — the true "how many different minifigures have ever existed" count, feeding the 19,082 figure.

'Average Parts per Set'

#,0.00

```
'Average Parts per Set' = AVERAGEX (inventories, DIVIDE ([Parts], [Sets]))
```

Average parts per set, computed per inventory then averaged — **DIVIDE** instead of / means it safely returns blank rather than an error when a set has zero parts.

04 · Insights

Three acts of the LEGO system

1940S-1970S · FOUNDATION

Establishing the system

12 sets (1940s) → 90 (1950s) → 172 (1960s) → 345 (1970s). Only Classic, City, Castle, and Space theme families exist — steady, linear growth, no diversification yet.

1980S-1999 • EXPANSION

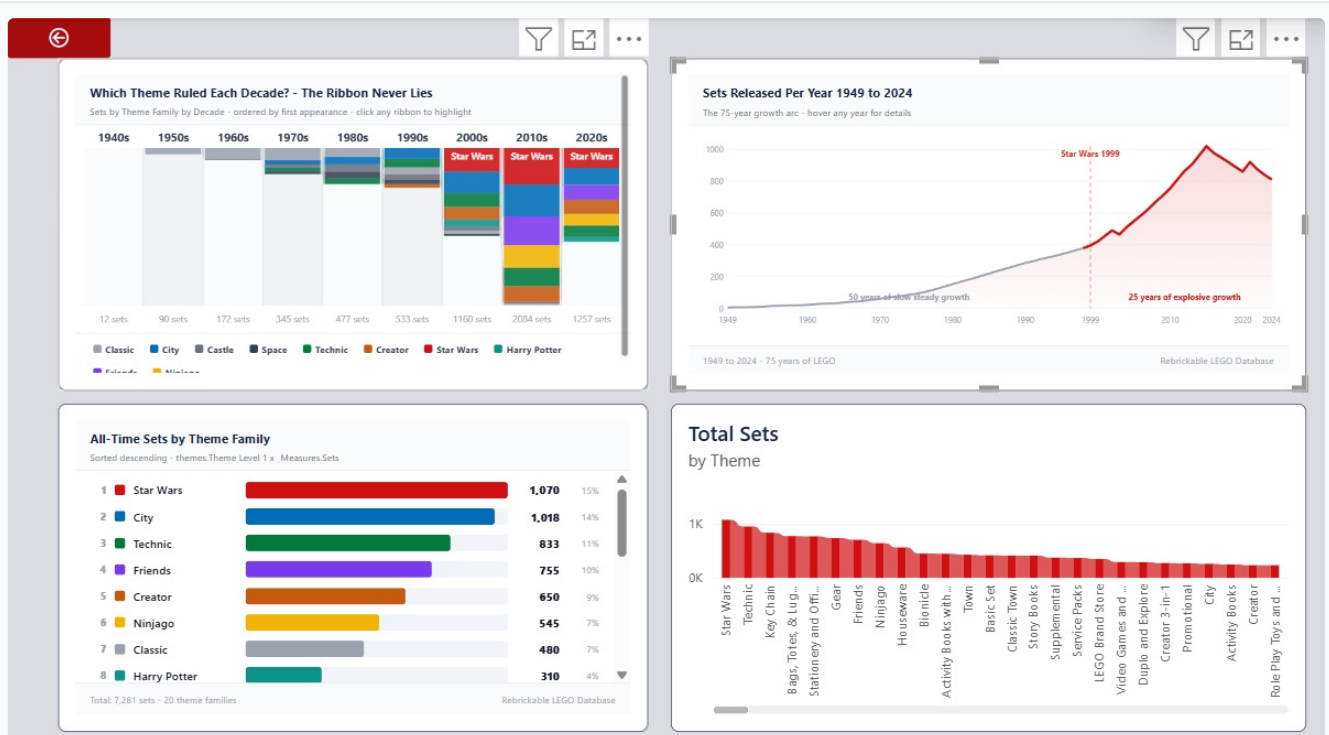
The inflection point

477 sets (1980s), 533 (1990s) — Technic and Creator join the lineup, but growth is still linear. Then 1999 hits: the Star Wars license launches, ending 50 years of slow, steady growth in a single year.

2000S-2024 • PLATFORM ERA

From toy to ecosystem

1,160 sets (2000s), peaking at 2,084 (2010s) — 25 years of explosive growth, with Harry Potter, Friends, and Ninjago all entering the theme roster. The 2020s show 1,257 sets so far, but that decade is only 5 years deep (2020–2024) — not yet a decline. Star Wars ends as the all-time #1 theme family (1,070 sets, 15%), ahead of City (1,018, 14%) and Technic (833, 11%), in the top-8 family rollup.



Source: Rebrickable LEGO Database · sets by theme family, 1949–2024.

Beyond the top 8 theme families, the long-tail chart shows sets spread across categories like Key Chain, Bags/Totes & Luggage, Gear, Houseware, Bionicle, Activity Books, Story Books, and Promotional items — the first hint that LEGO's growth wasn't just more core sets, it was the brick system extending into entirely new product categories.

THEME LIFECYCLE — WHO LASTS, WHO DISAPPEARS FAST



Top 10 longest-running themes (Promotional: 51 yrs; Seasonal & Technic: 50 yrs), 10 themes discontinued fastest (Monster Fighters and Dinosaurs: 1 year), and top 10 themes by total sets released (Duplo: 1,453; Star Wars: 1,094; City: 1,016).

The lifecycle view adds nuance the trend charts alone can't: Duplo, not Star Wars, is actually the single highest-volume theme ever (1,453 sets) — a reminder that the quiet, evergreen product can outproduce the headline licensed one. Meanwhile themes like Monster Fighters and Dinosaurs lasted just a single year, showing that not every experiment survives, even inside a system this stable.

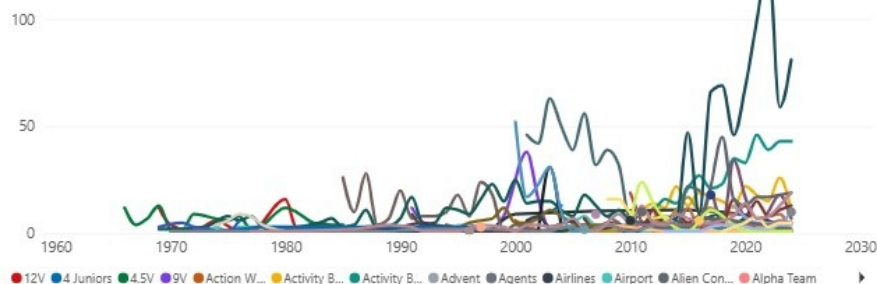
FROM BRICKS TO EVERYDAY MERCHANDISE

Products Life Span Shows the Real Transformation

From Simple Legos to Lego Sets to Every day merchandise

Total Sets

by Year and Theme



Year	img_url	Set
2022		Zyclops Chase
2008		ZX Turbo
2020		ZX 8000 LEGO Sneaker
2022		Zurg Battle
2014		Zorch

"Products Life Span Shows the Real Transformation" — total sets by year and theme; note the sharp proliferation of parallel theme lines from ~2020 onward, reflecting the shift toward everyday merchandise alongside core sets.

INTERACTIVE TIME CAPSULE (1949-2027)

LEGO® Time Capsule - The Story

STORY From 5 sets and 10 colors in 1949 to 27,069 sets, 273 colors, 149 themes and 19,082 minifigures by 2027 — 78 years of pure creativity.

THE BEGINNING - 1949

In 1949, a small Danish company called LEGO released just **5 sets** in **10 colors**. Nobody could have predicted that this wooden-brick toy company would grow into the **world's most valuable toy brand** — producing **27,069 sets** across **149 themes** and **273 colors** over the next 78 years.

27,069

TOTAL SETS EVER

Peak: 2025 with 1,339 sets in a single year

273

COLORS USED

From 10 in 1949: **83** still active today

19,082

MINIFIGURES

Born in 1975 - grew to 19,082 unique figures

1,094

STAR WARS SETS

Launched 1999 - peaked 2015 with 70 sets

THE TURNING POINTS

- 1958** — The iconic stud-and-tube brick is patented, making LEGO sets universally compatible.
- 1978** — The LEGO minifigure is born, transforming bricks into storytelling.
- 1999** — Star Wars arrives. 1,094 sets and counting — the most successful licensed theme in LEGO history.
- 2003** — Near bankruptcy. LEGO cuts themes, focuses on core products, and begins its greatest comeback.
- 2014** — The LEGO Movie grosses \$469M worldwide, reigniting the brand globally.
- 2021** — The World Map set launches with **11,695 pieces** — the largest set ever made.

The full arc: 5 sets/10 colors in 1949 → 27,069 sets, 273 colors, 149 themes, 19,082 minifigures. Turning points: 1958 stud-and-tube patent, 1978 minifigure introduced, 1999 Star Wars (1,094 sets and counting), 2003 near-bankruptcy and refocus, 2014 The LEGO Movie (\$469M), 2021 World Map set (11,695 pieces, the largest ever).

LEGO® Time Capsule

KEY TAKEAWAY From 10 colors and 5 sets in 1949, LEGO grew to 78 active colors, 142 themes, 24,814 sets, and 17,081 minifigures by 2024 — 78 years of creativity driven by licensed partnerships, fan demand, and relentless innovation.

1953 YEAR

COLORS

10

CUMULATIVE

6

✓ ACTIVE

4

✗ RETIRED

+ NEW ✗ DISCONTINUED

THEMES

1

CUMULATIVE

1

✓ ACTIVE

0

✗ RETIRED

+ NEW ✗ DISCONTINUED

SETS

15

CUMULATIVE

4

✓ ACTIVE

11

✗ RETIRED

+ NEW ✗ DISCONTINUED

MINIFIGS

0

CUMULATIVE

—

✓ ACTIVE

0

✗ RETIRED

+ NEW ✗ DISCONTINUED

COLORS

+0 new

ACCUMULATING PALETTE

NEW THIS YEAR

FULL CATALOG



THEMES

+0 new

ACCUMULATING CATALOG

NEW THIS YEAR



FULL CATALOG

System

SETS

+4 new

ACCUMULATING CATALOG

NEW THIS YEAR



FULL CATALOG

1949: +5 1950: +6 1953: +4

MINIFIGS

+0 new

ACCUMULATING COUNT

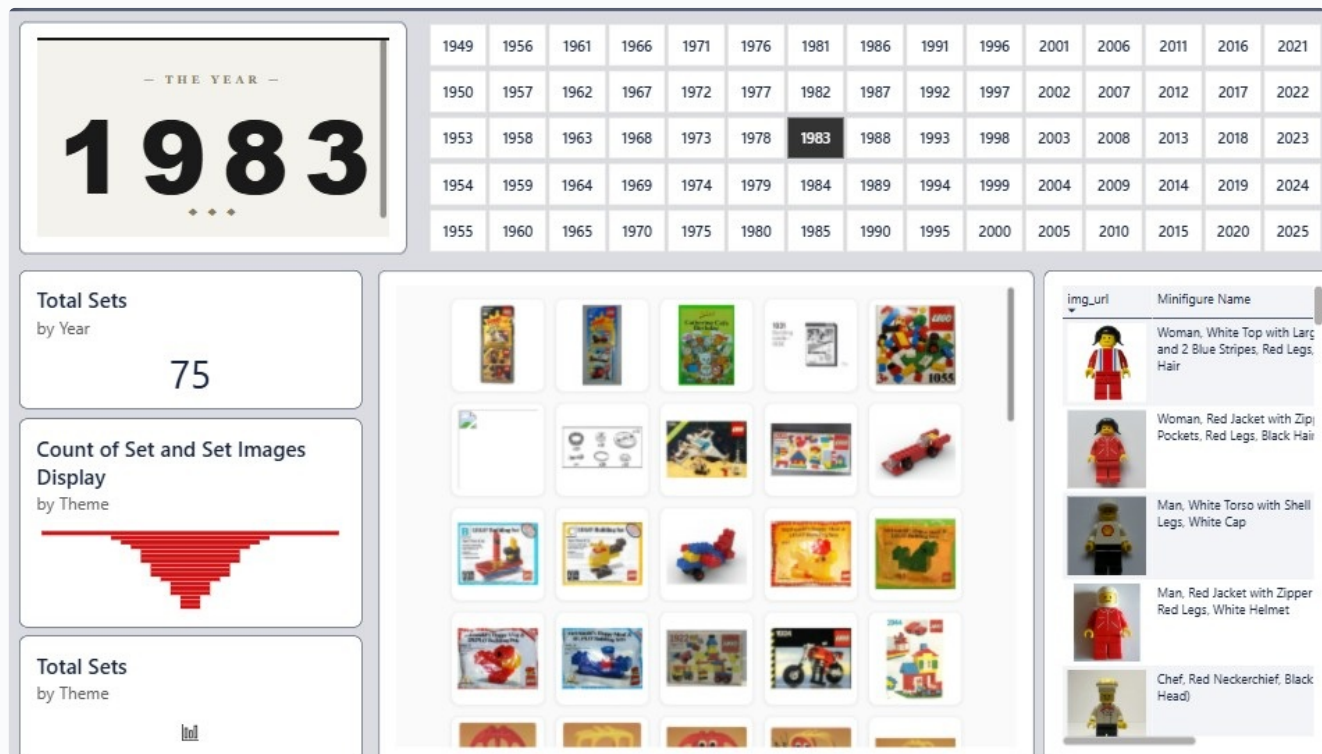
NEW THIS YEAR



MINIFIGURES BY YEAR

1949 1960 1970 1980 1990 2000 2010 2020 2027 **Pause** Speed: **Slow** **Normal** **Fast** **Faster** **Max** **Stop** **Pick Year** Source: (Redrickable LEGO Database (redrickable.com) - LEGO® is a trademark of the LEGO Group

Year-by-year explorer (shown at 1953): cumulative colors, themes, sets, and minifigures, each split into active vs. retired, with a play/pause slider spanning 1949–2027.



Drilling into a single year (1983): 75 sets released, with the actual set images and minifigure catalog surfaced alongside the trend — turning the dataset back into the physical products it represents.

RECONCILING THE NUMBERS

Different visuals in this report report different totals on purpose: the ribbon/ranked charts use a curated "top theme family" rollup (7,281 sets, 20 families) for readability, while the Time Capsule and merchandise-breakdown views report the full catalog (27,069 records, 149 themes) including non-brick merchandise. Both are correct — they answer different questions at different levels of the same model's Theme Level 1/2/3 hierarchy.

05 • Recommendations

The transferable takeaway

- **Keep the core stable, let the edges diversify** — LEGO's brick spec never changed since 1958; growth came from what was built on top of it.
- **Licensing as a proven growth lever** — the 1999 Star Wars launch is the single inflection point separating 50 years of slow, linear growth from 25 years of exponential output.
- **Merchandise is a second, quieter growth lever** — 28% of all catalog records are non-brick goods (keychains alone: 835). Treating adjacent merchandise as a deliberate category, not an afterthought, is a repeatable diversification pattern.
- **Evergreen themes can outperform headline licenses** — Duplo's 1,453 all-time sets beat Star Wars' 1,094, a reminder to protect steady, long-running product lines even while chasing licensed spikes.
- **Watch concentration and lifecycle risk together** — track both the top-3 theme-family concentration ratio (~40% of the curated rollup) and the discontinuation rate (61% of themes launched since 1949 have already ended) as paired portfolio-health signals.